Upstream Detector Placement: Interface and Space Questionnaire (for SAMURAI beamline integration)

Updated: 2026-03-06

1. Context

A dual-location detector idea was discussed (`upstream` and `in front of SAMURAI`). For this round, we only request **upstream-side** mechanical boundary inputs, so that interface sizing and installation feasibility can be fixed first.

This is a **primary-beam** experiment (not a secondary-beam setup).

2. Questions to Send (Upstream Only)

A. Upstream interface definition

1. What is the exact upstream interface standard (JIS / VG / VF / ICF / custom)?
2. What are the exact flange parameters at the upstream connection point?
   - nominal size (A)
   - ID / OD
   - bolt count and bolt-circle (PCD)
   - gasket/O-ring requirement
3. Is the mating side fixed as `VG` upstream and `VF` downstream for this section?
4. Is there any mandatory gate valve model/port standard at the upstream boundary?

B. Upstream installation space envelope

1. What is the available straight installation length along beam axis (mm)?
2. What is the allowed 3D envelope (X/Y/Z, mm) around the upstream installation location?
3. What is the maximum allowed mass and center-of-mass offset constraint?
4. Are there no-go volumes due to magnets, supports, shielding, cable trays, or maintenance paths?
5. What is the minimum service clearance needed for assembly/disassembly tools?

C. Upstream line-of-sight and aperture (geometry only)

1. Along upstream to SAMURAI-front path, what is the minimum effective aperture and where?
2. Are there fixed obstacles (slit/collimator/window/diagnostics/supports) that define the hard envelope?
3. Are there movable components that can intrude into this path during operation?

3. Reply Template (Upstream Only)

Item	Value	Notes
Interface standard		
Nominal size (A)		
Flange ID (mm)		
Flange OD (mm)		
Bolt circle PCD (mm)		
Bolt count		
O-ring / gasket type		
Available straight length (mm)		
Allowed envelope X (mm)		
Allowed envelope Y (mm)		
Allowed envelope Z (mm)		
Allowed mass (kg)		
Minimum effective aperture (mm)		

4. Beamline Figures and Fig.2 Pipe Specs

4.1 Newly added schematic (primary reference)

Source file: `schematic_of_beamline.pdf` (newly added by user).

4.2 Fig.2 and pipe-spec notes from `BeamLIneatSAMURAI.pptx`

The following notes are extracted from PPTX labels (slide text):

- GateValve: downstream side is `VF100`.
- Pipe (1): `VG100-VF150` converter.
- Pipe (2): `VG150-VF150` flexible pipe.
- Pipe (3): `VG150-VF150` duct.
- Guidance note in PPTX: connect to the gate valve standard first, then arrange the rest (JIS flange recommended).

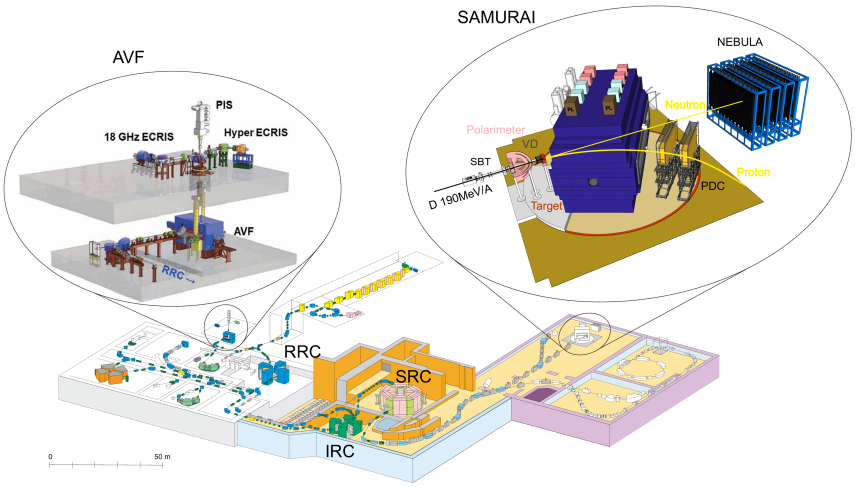


Figure 1: Beamline schematic (page 1)

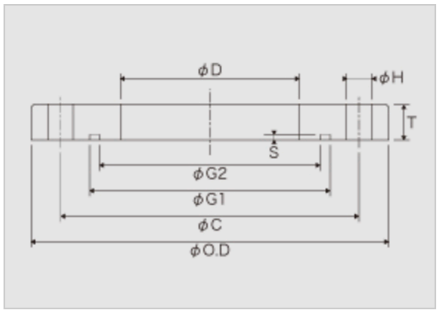
4.3 Quick flange mapping used in Fig.2 labels

Label in PPTX	Upstream side	Downstream side	Type note
GateValve	-	VF100	fixed gate valve boundary
(1) converter	VG100	VF150	converter
(2) flexible pipe	VG150	VF150	flexible section
(3) duct	VG150	VF150	rigid duct

5. Practical note

For this stage, only upstream interface and space constraints are requested.
Do not freeze final detector mechanics before Section 3 is fully answered.

VG 真空フランジ(JIS)



SS=Stainless Steel

材質記号	X	Z
材質	304 SS	316L SS

単位：(mm)

呼径	形式番号	適用パイプ D	O.D	厚さ	ボルト穴			ボルトの呼び	ガスケットのみぞ		
				T	C	N	φH		G1	G2	S
10A	VG-10	17.3	70	8	50	4	10	M8	34	24	3
20A	VG-20	27.2	80	8	60	4	10	M8	44	34	3
25A	VG-25	34.0	90	8	70	4	10	M8	50	40	3
40A	VG-40	48.6	105	10	85	4	10	M8	65	55	3
50A	VG-50	60.5	120	10	100	4	10	M8	80	70	3
65A	VG-65	76.3	145	10	120	4	12	M10	95	85	3
80A	VG-80	89.1	160	12	135	4	12	M10	110	100	3
100A	VG-100	114.3	185	12	160	8	12	M10	130	120	3
125A	VG-125	139.8	210	12	185	8	12	M10	160	150	3
150A	VG-150	165.2	235	12	210	8	12	M10	185	175	3
200A	VG-200	216.3	300	16	270	8	15	M12	241	225	4.5
250A	VG-250	267.4	350	16	320	12	15	M12	291	275	4.5
300A	VG-300	318.5	400	16	370	12	15	M12	341	325	4.5
350A	VG-350	355.6	450	20	420	12	15	M12	396	380	4.5
400A	VG-400	406.4	520	20	480	12	19	M16	446	430	4.5
450A	VG-450	457.2	575	20	553	16	19	M16	504	480	7
500A	VG-500	508.0	625	22	585	16	19	M16	554	530	7

Figure 2: Fig.2 related reference

VF 真空フランジ型(JIS)

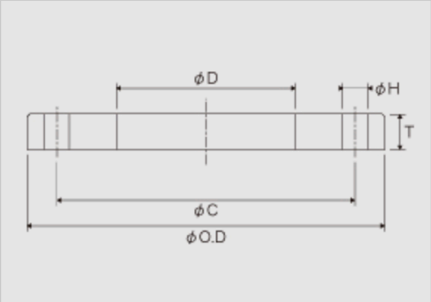


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Figure 3: Additional upstream/terminal-prep reference